Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/06/2010
PI Name:	Young, Laurence R. Sc.D.		
Project Title:	Sensorimotor Displays and Controls to Enhance the Safety of Human/Machine Cooperation During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lry@mit.edu	**Fax:**	FY 617-258-8111
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7759
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Department of Aeronautics and Astronautics		
PI Address 2:	77 Massachusetts Avenue		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	8
Comments:	Deceased as of August 2021.		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Oman, Charles (Massachusetts Institute of Technology) Wood, Scott (NASA JSC) Duda, Kevin (The Charles Stark Draper Laboratory, Inc.) Estrada, Arthur (US Army Aeromedical Research Laboratory)		
Grant/Contract No.:	NCC 9-58-SA01604		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Lunar landing depends on the selection and identification of an appropriate location that is level and free of hazards, along with a stable controlled descent to the surface. During crewed landings, astronauts are expected to interact with automated systems, based upon improved terrain maps and sensor updates, to perform tasks such as manual re-designation of landing point, adjustment of descent trajectory or direct manual control. However, sensorimotor limitations, both vestibular and visual, are likely to interfere with performance and safety. This integrated project examines the nature of the anticipated spatial disorientation and terrain perception limits as they affect the transition from automatic to manual control, and develops advanced display countermeasures to overcome these limitations. There are four specific aims investigated in this multi-institution effort: (1) Examine the nature of anticipated sensorimotor difficulties (e.g., spatial disorientation, limits on terrain perception) as they affect the transition from automatic to manual control, (2) Develop and evaluate advanced display countermeasures for enhancing situation and terrain awareness and for overcoming performance limitations caused by reduced visibility associated with lunar lighting, terrain reflectivity and the absence of atmosphere utilizing Draper Laboratory's fixed-base lunar lander cockpit simulator for full human-in-the-loop evaluation. (3) Evaluate the effectiveness of the cockpit displays during human-in-the-loop manual control in the JSC TTS during "critical" and "hover" tasks testing the tilt-translation and tilt-gain illusions of altered acceleration sensitivity as it applies to lunar gravity following a period of weightlessness, and (4) Perform a series of evaluations of the displays using the U.S. Army Aeromedical Research Laboratory's six-degree-of-freedom helicopter simulator as a lunar landing analog for replicating lunar lighting and the various parameters associated with dust "brownout" conditions. In our second year of work, we have made significant progress in Aims 1 and 2 and are preparing for Aim 3. We have analyzed a set of candidate Altair landing trajectories using a physiologically-based model of human spatial orientation and concluded that there is a likely risk of pilot perceptions differing from actual vehicle state. Since the first year, we have analyzed the candidate trajectories by incorporating the visual out-the-window-view portion of the physiologically-based model. Despite including visual cues, there is still a likely risk of pilot perceptions of orientation differing from that of the actual vehicle. Beyond the automatic candidate trajectories, landing point redesignation and manual control maneuvers have been analyzed. These vehicle maneuvers, which occur just prior to landing and while the pilot is actively in control, were also seen to likely create misperceptions in orientation. Within the model, we studied the effects of dust blowback, which was seen to have an impact if it occurs at higher altitudes or if vehicle maneuvers are made late in the landing. An experiment in conjunction with Dr. Bilimoria's work on handling qualities at NASA Ames Research Center was conducted. The study used the large motion base simulator, the Vertical Motion Simulator (VMS) to study human perceptions of orientation during lunar landing-like vehicle motions. This experiment studied both perceptions of vehicle roll and pitch angle as well as vehicle horizontal velocity. The study included test cases of vestibular only (blindfolded), visual out the window of a simulated lunar landscape, and visual on the display instruments. In Aim 2, we have developed an experimental testbed for lunar landing final approach and terminal descent in the Draper fixed-base simulator. This testbed simulation is being used to test novel display prototypes including an energy contour display, which provides the pilot information on which areas on the lunar surface are achievable given the remaining amount of fuel. In addition to the novel displays, a complete primary flight display (PFD) has been developed for simulation purposes. An experiment using these displays to land on the lunar surface is being carried out in the Draper simulator. Flight and landing performance, situation awareness, and pilot workload are being evaluated. In preparation for Aim 3, the Draper and MIT group has provided the NASA Johnson Space Center portion of the team with prototype displays for use in an experiment in the JSC Tilt-Translation Sled (TTS). Regarding Aim 4 experiments, which are not scheduled until Years 3-4, the dust simulation in the US Army 6-DOF helicopter simulation has been modified to account for the properties of lunar dust blowback. For the third year, we may continue our work in Aim 1 by performing a follow-on experiment in the Ames VMS or other motion-based simulator to further confirm and validate the physiologically-based model and the misperceptions seen during landing trajectories. In Aim 2, we will complete the experiment on the novel display prototype. Also we will refine this display prototype and continue development on others. The focus of Year 3 will be on the manual control experiment as described in the proposal to take place at NASA Johnson Space Center in the TTS. Further efforts will be made in preparation for the Year 4 work in the Army 6-DOF motion simulator. Other non-space examples of context-specific dual adaptation will be examined as well.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to determine the limits of human performance under likely landing conditions that may cause spatial disorientation. Appropriate roles can thereby be selected for humans and automated systems. This project will contribute to a better understanding of visual and vestibular conditions contributing to spatial disorientation during landing and the resulting effects on human manual control. We will have demonstrated display and control system interfaces to reduce pilot workload, improve situation awareness, and mitigate spatial disorientation to ensure a safe crewed lunar landing. Finally, the work may also have terrestrial applications in mitigating the risk of helicopter accidents by suggesting new techniques to address problems associated with brownout during landing. In particular the energy contour display being prototyped may have applications in helicopter flight planning, coping with brownout, as well as mission management aspects for guiding air-drops to locations given current environmental conditions (e.g. wind, air density).
Task Progress:	Progress for Aim 1: We have extracted relevant parameters from candidate trajectories within the NASA Autonomous Landing and Hazard Avoidance Technology (ALHAT) Project trade space. These parameters were used as inputs to the Observer Model, which is a physiologically based model of spatial orientation to estimate perceived orientation of the astronaut within the lunar landing vehicle from the initiation of the braking burn to descent from lunar orbit all the way through touchdown on the surface of the moon. In addition to the automated landings from ALHAT, landing point re-designation and manual control trajectories were studied as well. We had simulated the model both in the vestibular-only condition and also with visual out-the-window cues activated. Predictions of pilot perceptions were seen to differ from the actual vehicle orientation. The Observer model results were partially validated with a human subject experiment in the NASA-Ames VMS. In this experiment, subjects reported orientation perceptions during manually flown landings in conjunction with Dr. Bilimoria's lunar lander handling qualities experiment in the VMS. Progress for Aim 2: An experimental testbed for piloted lunar landing experiments has been developed in Draper's fixed base simulator. The simulator allows for different displays, control modes, and vehicle dynamics to be tested using human pilots interacting with the simulator through joystick hand controllers. The prototype control modes and primary flight displays that were implemented in MATLAB/Simulink for initial evaluation have been expanded upon and implemented in the Draper simulator. In addition to primary flight displays, novel display concepts have been developed.

Task Progress:	In particular a fuel-contour display, which provides the pilot information on achievable landing areas given the current fuel level, has been developed and implemented in the simulator for experimentation. Human subject experiments are just beginning focusing on pilot performance, situational awareness, and workload. Progress for Aim 3: The protocol for the Aim 3 experiment at Johnson Space Center (JSC) using the Tilt-Translation Sled (TTS) has been reviewed and approved by the JSC-IRB. We have worked towards integrating the display prototypes developed at Draper into the experimental protocol. The TTS has been modified to suit the Aim 3 experiment described in the proposal by adding a visual display fixed to the subject's chair. Progress for Aim 4: The US Army Aeromedical Research Lab has led an effort to augment the dust blowback simulation currently in the 6-DOF helicopter simulator. The dust blowback characteristics have been modified from those that occur during helicopter landing to those that are expected during lunar landings with regolith blowback. Dr. Estrada has also provided subject matter expertise on aeromedical/physiological issues, hover flight techniques, cockpit displays, and situational awareness.
Bibliography Type:	Description: (Last Updated: 01/21/2026)
Abstracts for Journals and Proceedings	Duda KR, Stimpson AJ, Clark TK, Young LR, Oman CM. "Flight displays and control modes for safe and precise lunar landing." 81st Annual Scientific Meeting of the Aerospace Medical Association, Phoenix, Ariz., May 9-13, 2010. Aviation, Space, and Environmental Medicine. 2010 Mar;81(3):327. , Mar-2010
Abstracts for Journals and Proceedings	Young LR, Duda KR, Clark TK, Stimpson AJ, Oman CM. "Sensorimotor interaction with vehicle displays and controls to enhance human-machine cooperation during lunar landing." 2010 NASA Human Research Program Investigators' Workshop, Houston, Tex., February 3-5, 2010. 2010 NASA Human Research Program Investigators' Workshop, Abstract Book, February 2010. , Feb-2010
Abstracts for Journals and Proceedings	Young LR, Duda KR, Oman CM, Liu AM, Stimpson AJ, Clark TK. "Critical factors affecting lunar landing supervisory control performance." 60th International Astronautical Congress, Daejeon, Republic of Korea, October 12-16, 2009. 60th International Astronautical Congress, Abstract Book, October 2009. , Oct-2009
Awards	Duda K. "AIAA Life Sciences and Systems Technical Committee, January 2010." Jan-2010
Dissertations and Theses	Clark TK. "Human spatial orientation perceptions during simulated lunar landing." Thesis, Massachusetts Institute of Technology, June 2010. , Jun-2010
Papers from Meeting Proceedings	Clark TK, Stimpson AJ, Young LR, Oman CR, Duda KR. "Analysis of human spatial perception during lunar landing." 2010 IEEE Aerospace Conference, Big Sky, MT, March 6-13, 2010. IEEE AC paper, 2010. http://dx.doi.org/10.1109/AERO.2010.5447026 , Mar-2010